

Assignment 2: Functions and Functionals

ME/BME 736 — Advanced Finite Element Methods

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Show the main steps in each calculation.

Problem 1 (10 marks): Equality almost everywhere

Define the measurable function $w : (0, 1) \rightarrow \mathbb{R}$ by

$$w(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

1. Use the fact that $\mathbb{Q} \cap (0, 1)$ is countable to show that $w = 0$ almost everywhere in $(0, 1)$.
2. Compute $\sup_{0 < x < 1} |w(x)|$ and $\text{ess sup}_{0 < x < 1} |w(x)|$.
3. Show that w and the zero function represent the same element of $L^p(0, 1)$ for every $1 \leq p \leq \infty$.

Problem 2 (10 marks): Uniform and L^2 convergence are different

We first recall the definition of convergence in a normed vector space.

Definition (convergence in a normed vector space). Let V be a normed vector space with norm $\|\cdot\|_V$, and let $(a_n)_{n \in \mathbb{N}}$ be a sequence in V . We say that $(a_n)_{n \in \mathbb{N}}$ converges to $a \in V$, or that a is the limit of the sequence in V , if

$$\|a_n - a\|_V \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Let $I = [0, 1]$ and define $u_n(x) = x^n$ for $x \in I$.

1. Determine the pointwise limit $u(x) = \lim_{n \rightarrow \infty} u_n(x)$ for every $x \in I$.
2. Show that $\|u_n - u\|_\infty = 1$ for every n . Hence determine whether u_n converges to u in the supremum norm.
3. Is the function u obtained in Part 1 also the limit of u_n in $L^2(I)$? Justify your answer. If it is, identify a constant function that is another representative of the same element of $L^2(I)$, and explain why it is equivalent to u in $L^2(I)$.

Problem 3 (10 marks): Extensions of Gauss divergence theorem

Let $\mathbf{v}$ and $\mathbf{a}$ be continuously differentiable vector fields, and let $\mathbf{A}$ be a continuously differentiable second-order tensor field. Using the scalar gradient theorem, prove each identity independently:

$$\begin{aligned}\int_{\Omega} \frac{\partial v_i}{\partial x_j} dx &= \int_{\partial\Omega} v_i n_j ds \quad (i, j = 1, \dots, d), \\ \int_{\Omega} \frac{\partial a_i}{\partial x_i} dx &= \int_{\partial\Omega} a_i n_i ds, \\ \int_{\Omega} \frac{\partial A_{ij}}{\partial x_j} dx &= \int_{\partial\Omega} A_{ij} n_j ds \quad (i = 1, \dots, d).\end{aligned}$$

Problem 4 (10 marks): Checking Gauss's theorem on a square

Let $\Omega = (0, 1)^2$ and $\mathbf{a} = (x_1^2 x_2, x_1 x_2^2)$. Compute $\int_{\Omega} \nabla \cdot \mathbf{a} dx$ directly. Then compute the outward flux on each side of $\partial\Omega$ and verify

$$\int_{\Omega} \nabla \cdot \mathbf{a} dx = \int_{\partial\Omega} \mathbf{a} \cdot \mathbf{n} ds.$$

Problem 5 (10 marks): Computing H^1 and H^2 norms

For $v(x) = x(1 - x)$ on $(0, 1)$, compute $\|v\|_{L^2(0,1)}$, $|v|_{H^1(0,1)}$, $\|v\|_{H^1(0,1)}$, $|v|_{H^2(0,1)}$, and $\|v\|_{H^2(0,1)}$.

Problem 6 (10 marks): Derivative of a nonlinear energy functional

For smooth scalar functions on a bounded domain and prescribed $k, f \in C(\overline{\Omega})$, let

$$J(u) = \int_{\Omega} \left(\frac{k}{2} |\nabla u|^2 + \frac{1}{4} u^4 - fu \right) dx.$$

Compute $DJ(u)[v]$ (also denoted by $\delta J(u; v)$) (see 2.9.4 in Chapter 2). Verify that the result is linear in v .

Problem 7 (15 marks): Continuity and coercivity of a diffusion form

Let $V = H_0^1(\Omega)$ with $\|v\|_V = \|v\|_{H^1(\Omega)}$, and let

$$a(u, v) = \int_{\Omega} k \nabla u \cdot \nabla v dx,$$

where $k \in L^\infty(\Omega)$ and

$$0 < k_0 \leq k(\mathbf{x}) \leq k_1 \quad \text{for almost every } \mathbf{x} \in \Omega.$$

Use the Cauchy–Schwarz and Poincaré inequalities (see **Reference inequalities** section in Chapter 2) to prove

$$|a(u, v)| \leq k_1 \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \quad \forall u, v \in V,$$

and

$$a(v, v) \geq \frac{k_0}{1 + C_P^2} \|v\|_{H^1(\Omega)}^2 \quad \forall v \in V.$$

Hence identify the continuity constant M and coercivity constant α ; see definitions of continuity and coercivity in section 2.10 of Chapter 2.

Problem 8 (25 marks): Weak derivative of a tent function

Let

$$u(x) = 1 - |x|, \quad x \in \Omega = (-1, 1).$$

1. Determine the classical derivative of u on $(-1, 0)$ and $(0, 1)$ and propose a weak derivative w on $(-1, 1)$.
2. Verify directly that w is the weak derivative of u . That is, show that

$$\int_{-1}^1 u(x) \phi'(x) \, dx = - \int_{-1}^1 w(x) \phi(x) \, dx$$

for every $\phi \in C_0^\infty(-1, 1)$ by splitting the integral at $x = 0$.

3. Use the definition of $H^1(-1, 1)$ to determine whether $u \in H^1(-1, 1)$.
4. Determine whether $u \in H^2(-1, 1)$. You will use the method of contradiction to answer this question. Follow the steps and for each step, answer the asked question by clearly writing your steps and assumptions.

Recall that if $u \in H^2(-1, 1)$, then its weak derivative w must belong to $H^1(-1, 1)$. Hence there must exist some $g \in L^2(-1, 1)$ such that

$$\int_{-1}^1 w(x) \phi'(x) \, dx = - \int_{-1}^1 g(x) \phi(x) \, dx \quad \forall \phi \in C_0^\infty(-1, 1).$$

Use the following steps to show that such a function g cannot exist.

Step 1. Using the piecewise form of w , split the integral at $x = 0$ and show that

$$\int_{-1}^1 w(x) \phi'(x) \, dx = 2\phi(0) \quad \forall \phi \in C_0^\infty(-1, 1)$$

Therefore, the condition for g to be a weak derivative of w reduces to

$$2\phi(0) = - \int_{-1}^1 g(x) \phi(x) \, dx \quad \forall \phi \in C_0^\infty(-1, 1). \quad (1)$$

Step 2. Choose any $\eta \in C_0^\infty(-1, 1)$ satisfying $\eta(0) = 1$. For $0 < \varepsilon < 1$, define

$$\phi_\varepsilon(x) = \eta\left(\frac{x}{\varepsilon}\right).$$

Then $\phi_\varepsilon(0) = 1$. Also,

$$\|\phi_\varepsilon\|_{L^2(-1,1)} = \sqrt{\varepsilon} \|\eta\|_{L^2(-1,1)}.$$

You do not need to prove this identity, but it follows directly by substituting the definition of ϕ_ε and using the change of variables $y = x/\varepsilon$.

Step 3. Apply the reduced weak-derivative identity to $\phi = \phi_\varepsilon$. Since $\phi_\varepsilon(0) = 1$, the LHS of Equation 1 is simply 2 for $\phi = \phi_\varepsilon$. Thus, Equation 1 is given by

$$2 = - \int_{-1}^1 g(x) \phi_\varepsilon(x) \, dx.$$

We take absolute value of the both side to get

$$2 = \left| \int_{-1}^1 g(x) \phi_\varepsilon(x) \, dx \right|. \quad (2)$$

Use the Cauchy–Schwarz inequality (see Section 2.11 of Chapter 2) on the above integral to show

$$2 \leq \|g\|_{L^2(-1,1)} \|\phi_\varepsilon\|_{L^2(-1,1)}.$$

Combine this with the estimate from Step 2 to obtain

$$2 \leq \|g\|_{L^2(-1,1)} \sqrt{\varepsilon} \|\eta\|_{L^2(-1,1)}.$$

Step 4. Let $\varepsilon \rightarrow 0$. Explain why the right-hand side tends to zero while the left-hand side remains equal to 2. Conclude that the resulting inequality is impossible.

Hence show that no $g \in L^2(-1, 1)$ can be the weak derivative of w (because we have shown it violates the integral relation in the definition of the weak derivative for some cases of test functions ϕ), and therefore determine whether $u \in H^2(-1, 1)$.